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Inventor(s) Chen; Xiaoming et al.

Roller device

Abstract

This application relates to a roller device. The device comprises: a roller; and a roller axle, the roller axle comprises: an axle portion configured to mount the roller such that the roller is rotatable about the axle portion; and a riveting portion at a mounting end of the roller axle with respect to an axial direction of the roller axle, and extending from the mounting end in a direction away from the axle portion, wherein, when the roller device is pressed against a carrier under force, the riveting portion can pierce the carrier such that the roller axle is riveted with the carrier through the riveting portion.

Inventors:	**Chen; Xiaoming (Jiangsu, CN), Huang; Zhiqiang (Jiangsu, CN), Xing; Wei (Jiangsu, CN), Duan; Dapeng (Jiangsu, CN)**
Applicant:	**PEM (CHINA) CO., LTD.** (Jiangsu, CN)
Family ID:	**1000008779350**
Assignee:	**PEM (CHINA) CO., LTD. (Jiangsu, CN)**
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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
2005/0111911	12/2004	Zdravkovic	403/283	B23P 19/062
2006/0251489	12/2005	Denham	411/34	F16B 37/068
2007/0104553	12/2006	Philipskotter	411/176	F16B 19/086
2009/0016656	12/2008	Blair	29/898.1	F16C 13/006
2010/0232906	12/2009	Singh	411/501	B21J 15/36
2012/0180305	12/2011	Bartig	411/501	F16B 19/086
2013/0294866	12/2012	Singh	411/501	B21J 15/36
2016/0238080	12/2015	Wang	N/A	F16C 13/006
2016/0332215	12/2015	Blacket	N/A	B23K 20/24
2017/0343026	12/2016	Schaeffer	N/A	F16B 13/066
2017/0343027	12/2016	Wissling	N/A	B21J 15/36
2018/0080096	12/2017	Van Hall	N/A	F16B 19/086
2018/0117666	12/2017	Trinick	N/A	B21K 1/60
2018/0266465	12/2017	Heger	N/A	B21J 15/025
2019/0353202	12/2018	Wang	N/A	F16C 33/6696
2020/0156143	12/2019	Kawasaki	N/A	B21J 15/36
2021/0207638	12/2020	Bartig	N/A	F16B 19/086

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
1132391	12/1995	CN	N/A
2330235	12/1998	CN	N/A
202914521	12/2012	CN	N/A
203938315	12/2013	CN	N/A
204234651	12/2014	CN	N/A
209073800	12/2018	CN	N/A
2552372	12/1984	FR	N/A

OTHER PUBLICATIONS

International Search Report of PCT/CN2021/122758, mailed on Dec. 16, 2021. cited by applicant

Primary Examiner: Cook; Kyle A

Attorney, Agent or Firm: Junhe Law Office P.C.

Background/Summary

TECHNICAL FIELD
(1) The present disclosure relates to the field of riveting technologies, and particularly relates to a roller device.
BACKGROUND
(2) Some roller devices need to be mounted on a mobile rack, a cabinet or a similar mobile device to improve the mobility of these mobile devices. However, the existing roller devices are difficult to mount on the mobile device, and holes often need to be punched on the mobile device in advance for mounting, which increases the mounting difficulty and production cost.
(3) Therefore, it is desired to provide a roller device convenient for mounting.
SUMMARY
(4) An objective of the present application is to provide a roller device convenient for mounting.
(5) According to an aspect of the application, a roller device is provided, the device comprises: a roller; and a roller axle, the roller axle comprises: an axle portion configured to mount the roller such that the roller is rotatable about the axle portion; and a riveting portion at a mounting end of the roller axle with respect to an axial direction of the roller axle, and extending from the mounting end in a direction away from the axle portion, wherein, when the roller device is pressed against a carrier under force, the riveting portion can pierce the carrier such that the roller axle is riveted with the carrier through the riveting portion.
(6) In some embodiments of the application, the riveting portion and the axle portion do not overlap with each other in the axial direction of the roller axle.
(7) In some embodiments of the application, the roller axle further comprises a flange between the axle portion and the riveting portion, wherein the flange is configured to prevent the roller from contacting the carrier when the roller device is riveted with the carrier.
(8) In some embodiments of the application, the axle portion comprises an annular positioning mechanism extending around the axle portion, wherein the annular positioning mechanism is configured to prevent the roller from moving in the axial direction.
(9) In some embodiments of the application, the annular positioning mechanism protrudes from a surface of the axle portion.
(10) In some embodiments of the application, the annular positioning mechanism is recessed from a surface of the axle portion.
(11) In some embodiments of the application, the annular positioning mechanism is aligned with an axially central position of the roller.
(12) In some embodiments of the application, the roller is pre-mounted on the roller axle.
(13) In some embodiments of the application, the axle portion extends at least beyond the roller in the axial direction of the roller axle.
(14) In some embodiments of the application, the riveting portion comprises at its free end a wedge-shaped section which is substantially parallel to the axial direction of the roller axle, wherein the wedge-shaped section flares when the roller device is pressed against the carrier.
(15) In some embodiments of the application, the riveting portion comprises at its free end a

mounting section which is perpendicular to the axial direction of the roller axle, wherein an outer periphery of the mounting section has a regular polygonal shape.
(16) In some embodiments of the application, the mounting section has an inner periphery of a regular polygonal shape, wherein each face of the inner periphery is parallel to a face of the outer periphery.
(17) In some embodiments of the application, the mounting section has an inner periphery of a circular shape.
(18) In some embodiments of the application, the riveting portion comprises at its free end a mounting section which is perpendicular to the axial direction of the roller axle, and wherein the mounting section has an outer periphery of a circular shape.
(19) The foregoing is an overview of the present application, which may simplify, summarize, and omit details. Those skilled in the art will appreciate that this section is merely illustrative and not intended to limit the scope of the present application in any way. This summary section is neither intended to identify key features or essential features of the claimed subject matter nor intended to act as an auxiliary means for determining the scope of the claimed subject matter.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) The foregoing and other features of the present application will be more fully understood from the following description and the appended claims taken in conjunction with the accompanying drawings. It is to be understood that these drawings depict only a few embodiments of the contents of the present application and should not be construed as limiting the scope of the present application. The contents of the present application will be illustrated more clearly and in more detail with the accompanying drawings.
(2) FIG. **1** depicts a roller device **100** according to an embodiment of the present application.
(3) FIG. **2** to FIG. **4** depict a process of mounting the roller device **100** shown in FIG. **1** on a carrier by riveting.
(4) FIG. **5** depicts an enlarged view of the roller device **100** shown in FIG. **1**.
(5) FIG. **6** depicts a perspective view of a riveting portion in the roller device according to an embodiment of the present application.
DETAILED DESCRIPTION
(6) In the following detailed description, reference is made to the accompanying drawings which form a part hereof. In the drawings, similar reference numerals generally refer to similar parts unless the context clearly dictates otherwise. The illustrative embodiments described in the detailed description, drawings and claims are not intended to be limiting. Other embodiments may be employed, and other changes may be made without departing from the spirit or scope of the subject matter of the present application. It is to be understood that various configurations, substitutions, combinations, and designs of the various forms of the present application, which are generally described in this application and are illustrated in the drawings, are intended to constitute a part of the present application.
(7) FIG. **1** depicts a roller device **100** according to an embodiment of the present application. In some embodiments, the roller device **100** can be mounted on a carrier, such as a metal plate, or other similar structures that can undergo plastic deformation.
(8) As shown in FIG. **1**, the roller device **100** includes a roller **102**, which can be made of rubber, plastic, metal or other materials. The roller device **100** also includes a roller axle **104**. The roller **102** is mounted on the roller axle **104**, so that the roller **102** is rotatable about an axle portion **106** of the roller axle **104**. In some embodiments, the roller **102** can be pre-mounted on the roller axle **104**, so that the roller device **100** can be provided as an integrated assembly. In other embodiments,

the roller **102** can also be provided separately from the roller axle **104**, which can be mounted on the roller axle **104** before or after the roller device **100** is mounted on the carrier.

(9) The axle portion **106** is configured as an extension structure with a central axis **108**, which is coaxial with a rotational axis of the roller **102**. In some embodiments, the axle portion **106** extends at least beyond the roller **102** in the axial direction of the roller axle **104**. In other words, the axial length of the axle portion **106** is greater than that of the roller **102**. In some other embodiments, the axial length of the roller **102** may also be shorter than or equal to that of the axle portion **106**.

(10) In some embodiments, the axle portion **106** has an annular positioning mechanism **110**, which can be arranged approximately near a central position of the axle portion **106**. Preferably, the annular positioning mechanism **110** can be aligned with the axially central position of the roller **102**. The annular positioning mechanism **110** may extend a certain angle around the axle portion **106**, such as 360 degrees or less. In some embodiments, the annular positioning mechanism **110** may be a continuous and complete ring, that is, extend 360 degrees, or may also include a plurality of separated positioning segments arranged around the axle portion. The annular positioning mechanism **110** can cooperate with the roller **102** to prevent the roller from moving on the axle portion **106** in an axial direction. In some embodiments, the annular positioning mechanism **110** can protrude from the surface of the axle portion **106**. Accordingly, the inner periphery of the roller **102** can be provided with an annular groove **112** matching with the annular positioning mechanism **110**. In other alternative embodiments, the annular positioning mechanism **110** may be recessed from the surface of the axle portion **106**. Accordingly, the inner periphery of the roller **102** can be provided with an annular protrusion matching with the annular positioning mechanism **110**. In some embodiments, the axle portion **106** may include a plurality of annular positioning mechanisms **110**, which are spaced in the axial direction on the axle portion **106**. The plurality of annular positioning mechanisms **110** can better prevent the roller **102** from moving in the axial direction. In some embodiments, the axial movement of the roller **102** can also be prevented by other positioning structures. For example, a pair of flanges (not shown) can be formed on the axle portion and at two sides of the roller, and the distance of the pair of flanges in the axial direction is roughly equal to or slightly greater than an axial length of the inner periphery of the roller. The protruded height of the pair of flanges from the axle portion can be greater than or equal to the thickness of the roller in a radial direction, or shorter than the thickness of the roller in the radial direction.

(11) The roller axle **104** also includes a riveting portion **114**, which is located at a mounting end **116** of the roller axle **104** with respect to an axial direction of the roller axle. The mounting end **116** is one of the two ends of the roller axle **104** in the axial direction away from the roller **102**. The riveting portion **114** may extend from the mounting end **116** in a direction away from the shaft portion **106**. When the roller device **100** is pressed against a carrier under force, the riveting portion **114** can pierce the carrier so that the roller axle **104** is riveted with the carrier through the riveting portion **114**. In some embodiments, the riveting portion **114** has a substantially cylindrical shape, and the riveting portion **114** includes a wedge-shaped section **118** at its free end, which is substantially parallel to the axial direction of the roller axle **104**. The wedge-shaped section **118** is configured to flare, when the roller device **100** is pressed against the carrier, so as to realize the riveting between the roller device **100** and the carrier. In some embodiments, the wedge-shaped section **118** can have an outward sloping surface. When the roller device **100** is being mounted, the sloping surface can resolve the pressure force into a radial outward force partially, so that the riveting portion **114** can fully flare.

(12) In some embodiments, the riveting portion **114** and the axle portion **106** do not overlap with each other in the axial direction of the roller axle **104**, that is, the roller **102** cannot be mounted on the riveting portion **114**. Since the riveting portion **114** may deform during the mounting of the roller device **100**, the riveting portion **114** and the roller **102** relatively separated can avoid the limited rolling performance of the roller **102** due to the deformation of the riveting portion **114**

(especially the expansion) after mounting. In some embodiments, the roller axle **104** also includes a flange **120** located between the axle portion **106** and the riveting portion **114**, which is used to prevent the roller **102** from contacting the carrier when the roller device **100** is riveted with the carrier.

(13) FIG. **2** to FIG. **4** depict a process of mounting the roller device **100** shown in FIG. **1** on a carrier by riveting.

(14) As shown in FIG. **2**, the mounting process starts with placing the roller device **100** on a side of the carrier **130**. In some embodiments, the carrier **130** can be provided with preformed through-holes or depressions (not shown in the figure), whose section shape and size on a plane perpendicular to the axial direction of the roller axle **104** substantially correspond to the shape and size of the end of the riveting portion **114**. Providing the through-holes or depressions improves the accuracy of the mounting of the roller device **100**. However, it can be understood that the roller device **100** according to an embodiment of the present application does not require through-holes or depressions to be provided on the carrier **130** in advance. As shown in FIG. **2**, the carrier **130** can generally be in the form of a flat structure. Since it is not desired to arrange the through-holes or depressions on the carrier **130** as guiding structures for mounting, the roller device **100** can be mounted on the carrier of various shapes, which greatly enlarges the application of the roller device **100**.

(15) In order to mount the roller device **100**, an upper loading tool **140** and a lower supporting tool **150** are respectively arranged on both sides of the roller **102** and the carrier **130**, which can move relatively to provide a pressure force to the roller device **100**. When mounting the roller device **100**, since the riveting portion **114** of the roller device **100** can pierce the carrier **130** and penetrate into the inside of the carrier **130**, the lower support tool **150** does not need to be preformed as a mold for the riveting portion **114**, but only needs to provide a supporting plane on a side close to the carrier **130**. On the other hand, during the mounting process, the upper loading tool **140** can contact a supporting end **122** of the roller axle **104**, which is opposite to the mounting end **116**, and apply a pressure force to the roller axle **104** through the supporting end **122**. In the example shown in FIG. **2**, the supporting end **122** of the roller axle **104** extends beyond the roller **102**, so the roller **102** may not be subjected to the pressure force applied by the upper loading tool **140** during mounting, which can avoid an axial movement of the roller **102** on the roller axle **104**.

(16) Still referring to FIG. **3**, the upper loading tool **140** and the lower supporting tool **150** move relatively and apply a pressure force to the roller axle **104**, which makes the riveting portion **114** continue to enter the inside of the carrier **130**. At this time, the riveting portion **114** undergoes plastic deformation and flares under force, and there is a relatively maximum outward expansion distance at its end, so that it is riveted in the carrier **130**. It can be seen that during the mounting process, the flange **120** can limit a depth of the roller axle **104** entering the carrier **130**, thus acting as a stop member for mounting.

(17) Referring to FIG. **4**, after mounting, the upper loading tool and the lower supporting tool can be removed, and only the carrier **130** and the roller device **100** that are riveted to each other are retained. The carrier portion **132** inside the riveting portion **114** can remain on the carrier **130** after mounting, which can improve the riveting strength and protect the riveting portion **114**.

(18) In combination with FIG. **2** and FIG. **3**, at the beginning of the mounting of the roller device **100**, the riveting portion **114** of the roller device **100** applies a force on the carrier **130**, which acts as a blanking tool to form riveting holes on the carrier **130**. Then, as the riveting portion **114** continues to move towards the carrier **130** under the force, depending on the initial shape of the riveting portion **114**, the carrier **130** will apply a reaction force on the riveting portion **114**, which causes the riveting portion **114** to produce expanded plastic deformation, thus forming a solid mechanical interlock structure between the roller device **100** and the carrier **130**. This self-punching does not require punching on the carrier in advance, so its mounting process has substantially no damage to the carrier with coating or cladding on the surface, which can improve a

service life of the mechanical structure after mounting.

(19) FIG. **5** depicts an enlarged view of the roller device **100** shown in FIG. **1**. As previously described in combination with FIG. **2** and FIG. **3**, the initial shape and structure of the riveting portion have an important impact on the strength after mounting. In the embodiment shown in FIG. **5**, the wedge-shaped section at the end of the riveting portion includes an inclined plane close to the inside of the riveting portion. In some embodiments, the inclined plane has an angle of about 35 to 80 degrees (relative to a surface plane close to a mounting position of the carrier), preferably 45 to 70 degrees, and more preferably 45 to 60 degrees. At the outermost side of the inclined plane along the axial direction, that is, the position where the initial contact with a mounting surface of the carrier happens when mounting the roller device, the wedge-shaped section can have a contact end surface with a radial width (parameter D in FIG. **5**), which is, for example, 0-0.5 mm, preferably 0.1 to 0.3 mm. The existence of the radial width can improve the mechanical strength of the end of the riveting portion, but may not significantly increase a resistance of the riveting portion when piercing into the carrier.

(20) In the example shown in FIG. **5**, the end of the riveting portion is also provided with a fillet surface, which is located at the outermost side of the riveting portion in the radial direction. The fillet surface and the inclined surface clamp the contact end surface between them, but the fillet surface can have a smaller radial width compared with the radial width C of the inclined surface (in FIG. **5**, it is represented as the difference between the half of the difference between the outer diameter and the inner diameter of the riveting portion (A−B)/2 and the radial width (C+D)), so that after mounting, radial forces of the riveting portion of the roller device are unbalanced, and the riveting portion flares as a whole.

(21) In order to firmly fix the riveting portion in the carrier, the riveting portion also has a predetermined axial length. It can be understood that the axial length depends on the thickness of the carrier on the one hand, and on the plastic deformation of the riveting portion after mounting on the other hand. In some embodiments, the axial length L**1** of the riveting portion (starting from a side of the flange close to the carrier in FIG. **5**) is substantially equal to the sum of the thickness of the carrier and 10% to 50% of the outer diameter A of the riveting portion, preferably equal to the sum of the thickness of the carrier and 20% to 40% of the outer diameter A of the riveting portion. This parameter configuration can allow the riveting portion to substantially expose from a non-mounting surface of the carrier or basically to reach the non-mounting surface of the carrier after mounting. It can be understood that in some other embodiments, the riveting portion can also have an axial length smaller than the thickness of the carrier. In the example shown in FIG. **5**, an arc transition can be provided between the riveting portion and the flange, and its axial length to the end of the riveting portion is L**2**.

(22) The mechanical connection strength of the roller device shown in the embodiment of this application is significantly better than that of the roller device with a common riveting structure or similar connectors. According to the inventor's test results, when mounting on a metal plate of the same thickness and material, in order to achieve the same connection strength, the outer diameter of the riveting portion of the roller device according to an embodiment of the present application can be half of that of the existing common riveting connection structure (the riveting portion does not flare significantly).

(23) FIG. **6** depicts a side view of a riveting portion in the roller device according to an embodiment of the present application. In some embodiments, the riveting portion has a mounting section at its free end that is perpendicular to the axial direction of the roller axle. That is, the mounting section is substantially parallel to the carrier to be mounted. In some embodiments, an outer periphery of the mounting section has a regular polygonal shape. In some examples, an inner periphery of the mounting section have a circular shape, as shown in FIG. **6**. In other examples, the inner periphery of the mounting section can have a regular polygonal shape, and each face of the inner periphery is parallel to a face of the outer periphery. Alternatively, the outer periphery of the

mounting section can also have a circular shape, that is, the riveting portion has a substantially cylindrical shape.

(24) It should be noted that although several modules or submodules for the roller device have been mentioned in the above detailed description, such division is exemplary and not mandatory. Practically, according to the embodiments of the present application, the features and functions of two or more modules described above can be embodied into one module. In contrast, the features and functions of a module described above can be further divided into multiple modules to be embodied.

(25) Those skilled in the art can understand and implement other variations to the disclosed embodiments from a study of the specification, the disclosure and accompanying drawings, and the appended claims. In the claims, the word “comprising” does not exclude other elements or steps, and the indefinite article “a” or “an” does not exclude a plurality. In applications according to present application, one element may conduct functions of several technical feature recited in claims. Any reference numerals of the drawings in the claims should not be construed as limiting the scope.

Claims

1. A roller device, the device comprising: a roller; and a roller axle, the roller axle comprising: an axle portion configured to mount the roller such that the roller is rotatable about the axle portion; and a riveting portion at a mounting end of the roller axle, wherein the mounting end of the roller axle is one of the two ends of the roller axle along an axial direction of the roller axle, and the riveting portion extends from the mounting end in a direction away from the axle portion, wherein the riveting portion comprises, at its free end, an inclined surface, a contact end surface, and a fillet surface extending from the inside to the outside, and the radial width of the fillet surface is smaller than the radial width of the inclined surface; and wherein, when the roller device is pressed against a carrier under force, the riveting portion can pierce the carrier such that the roller axle is riveted with the carrier through the riveting portion; wherein the riveting portion and the axle portion do not overlap with each other in the axial direction of the roller axle; wherein the axle portion comprises an annular positioning mechanism extending around the axle portion, wherein the annular positioning mechanism is configured to prevent the roller from moving in the axial direction.
2. The roller device of claim 1, wherein the roller axle further comprises a flange between the axle portion and the riveting portion, wherein the flange is configured to prevent the roller from contacting the carrier when the roller device is riveted with the carrier.
3. The roller device of claim 1, wherein the annular positioning mechanism protrudes from a surface of the axle portion.
4. The roller device of claim 1, wherein the annular positioning mechanism is recessed from a surface of the axle portion.
5. The roller device of claim 1, wherein the axle portion extends at least beyond the roller in the axial direction of the roller axle.
6. The roller device of claim 1, wherein the free end of the riveting portion forms a wedge-shaped section which is substantially parallel to the axial direction of the roller axle, wherein the wedge-shaped section flares when the roller device is pressed against the carrier.
7. The roller device of claim 1, wherein the free end of the riveting portion forms a mounting section which is perpendicular to the axial direction of the roller axle, and wherein the mounting section has an outer periphery of a circular shape.
8. The roller device of claim 1, wherein the annular positioning mechanism is aligned with an axially central position of the roller.
9. The roller device of claim 8, wherein the roller is pre-mounted on the roller axle.

10. The roller device of claim 1, wherein the free end of the riveting portion forms a mounting section which is perpendicular to the axial direction of the roller axle, wherein an outer periphery of the mounting section has a regular polygonal shape.

11. The roller device of claim 10, wherein the mounting section has an inner periphery of a regular polygonal shape, wherein each face of the inner periphery is parallel to a face of the outer periphery.

12. The roller device of claim 10, wherein the mounting section has an inner periphery of a circular shape.